

Setup Guide

Real-Time implementation of the NLMS Algorithm - EE310

Overview

This guide is for checkpoint 4, covering how to set up and use the physical implementation of the NLMS algorithm, divided into phases.

Hardware Setup

If using Windows 10/11, go to settings->system->sound->properties. Select the device's speaker and in the menu that opens, disable Audio enhancements. Do the same for the device's microphone.

Run tests to ensure the microphone and speaker are working.

Important: restart your computer before running the code. There have been instances where residual memory and background processes considerably worsened the results of the algorithm. For optimal performance, please restart and keep as few applications open as possible.

Necessary Libraries

Run the following in the VScode terminal:

```
pip install numpy PyAudio matplotlib ipython mutagen
```

This will install the necessary libraries used in the code onto your device.

Running Code

Run the file called 'realtime_demo.m'.

This will generate a series of labelled plots for a bunch of phases.

Details for each phase:

Phase 0, Baseline: Run the system with adaptation disabled (weights fixed at zero). We should hear the raw echo in the room. This confirms that echo is present and that your system is not relying on OS-level cancellation.

Phase 1, Convergence: Enable adaptation and remain silent. The echo should fade noticeably within a few seconds as the filter converges.

Phase 2, Double-Talk: While the far-end is playing and the filter is converged, speak naturally into the microphone. Your voice should be audible at the output while the echo remains suppressed.

Phase 3, Path Change: While the system is converged, physically change the acoustic environment, that is, place a book in front of the speaker, rotate the laptop, or cover the microphone partially and then uncover it. You will hear a brief burst of residual echo as the old filter weights become invalid, followed by re-convergence to the new path.